GEK-83809

DRIVE SYSTEM INSTRUCTIONS

POWER SYSTEM STABILIZER,

3S7932LA202G14

Includes speed transducer for toothed wheel, magnetic pickup, and torsional filter.

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

(GEK-83809)

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The information herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Department, USA.

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INTRODUCTION

The Power System Stabilizer (PSS) is designed to introduce a supplementary control signal into continuously acting voltage regulator equipment. The equipment will supplement the conventional voltage regulator control with a control function derived so that, under deviations of generator speed or load, excitation becomes controlled as a composite function of voltage and shaft speed. Under certain conditions, the composite function has been shown to be an aid to power system stability.

The PSS receives a signal from a toothed wheel with magnetic pick-up located at the turbine front standard. This signal, which is frequency modulated, is passed through a speed transducer which then has an output frequency equal to the modulation frequency and an amplitude proportional to the deviation in frequency. This signal is then passed through two lead-lag stages in order to obtain the proper phase shift characteristics, a band reject filter to attenuate signals at torsional frequencies, a washout stage to eliminate the effect of component drift and slow system frequency variations, a gain stage to obtain maximum control action and an output stage which interfaces the PSS with continuously acting voltage regulating equipment.

The Power System Stabilizer includes the functions of input transducer (not including the toothed-wheel magnetic pickup), signal conditioning, limiting, output circuitry, and attenuation of torsional frequencies. Figure 13 is a typical functional elementary diagram.

The Power System Stabilizer will apply the following transfer function to modify the output of the speed transducer for insertion into the voltage regulator:

$$G(s) = \frac{K\,(T_1 S)\,(1 + T_2 S)\,(1 + T_4 S)}{(1 + T_1 S)\,(1 + T_3 S)\,(1 + T_5 S)}$$

Adjustments are provided to vary the parameters of the PSS so as to permit compensation for variations in the characteristics of the regulator, excitation system, generator and power system to which it is applied. These adjustments can be made by calibrated dial settings which will be accurate to within 10% of the desired parameter value. The adjustments are capable of modifying the transfer function parameters through the following typical ranges:

K = 1 to 100

T_1 = 0.1 to 50 seconds

T_2 = 0.2 to 1.5 seconds

T_3 = 0.02 to 0.1 seconds

T_4 = 0.2 to 1.5 seconds

T_5 = 0.02 to 0.1 seconds

RECEIVING AND HANDLING

The Power System Stabilizer is normally supplied mounted in either the regulator or exciter cubicle. The instruction manual for the particular excitation system used should be consulted for receiving and handling instructions. If the PSS is shipped separately (as in the case of a retrofit system), then immediately upon receipt, the equipment should be carefully unpacked to avoid damage. It should be examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric sales office should be notified promptly.

STORAGE

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected from accidental damage. Particular care should be exercised to avoid storing the equipment in locations where construction work is in progress. If equipment is to be stored for more than three months, it should be kept heated to at least 5°C above ambient temperature to prevent condensation.

ABBREVIATIONS

The following abbreviations are used in this instruction manual:

ac	Alternating Current
Amp	Amperes
Db	Decibels
dc	Direct Current
FET	Field Effect Transistor
Hz	Hertz
IC	Integrated Circuit'
LED	Light Emitting Diode
Ma	Milliampere
Mv	Millivolt
PP	Peak-to-Peak
PSS	Power System Stabilizer
SCR	Silicon Controlled Rectifier
VAC	Alternating Current Voltage
VDC	Direct Current Voltage

DESCRIPTION (COMPONENTS)

The Power System Stabilizer consists of:

3S7932LA202G14 Panel

The Power System Stabilizer is normally mounted in either the regulator cubicle or exciter cubicle and contains the following:

NUMBER	NAME	QUANTITY
44C372607-G01	4800 Hz Speed Transducer	1
44C331857-G01	Signal Conditioning Board	1
44C331859-G01	Washout and Output Board	1
44C331863-G02	Relay Board	1
44C331861-G01	Ceiling Sensing Board	1
44C372640-G01	Band Reject Filter	1
44C331862-G01	Power Supply Board	1
3S7932MA321G9	Power Supply Panel(±36VDC)	1
3S7932MA321G10	Power Supply Panel (24VDC)	1

INSTALLATION

Location and Connections

The Power System Stabilizer is normally furnished as an integral part of the excitation and regulation equipment. (Refer to the drawings furnished with the excitation system and the instruction manual for the excitation system for location and connection instructions.)

Initial Checks

1. Wire check all external connections to the Power System Stabilizer. Be sure that wire size and type are adequate for the expected signal levels. (Refer to the connection diagram furnished with the excitation equipment.)

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DO NOT HI-POT OR MEGGER ANY CIRCUIT, AS THIS MAY RESULT IN DAMAGE TO COMPONENTS. ANY WIRING OR CIRCUITS TO BE HI-POTTED OR MEGGERED MUST FIRST BE DISCONNECTED FROM THE PSS.

2. Insure that all printed circuit boards are fully inserted.

3. Insure that the shield on the cable from the magnetic pick-up is grounded only at the excitation equipment.

OPERATION

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MISOPERATION OF THE POWER SYSTEM STABILIZER WHILE CONNECTED TO THE EXCITATION SYSTEM MAY RESULT IN DAMAGE TO THE GENERATOR. DEVIATIONS FROM THE FOLLOWING OPERATIONAL CHECKS AND PROCEDURES SHOULD BE CONSIDERED IN LIGHT OF THIS AND SHOULD BE DISCUSSED WITH THE MANUFACTURER PRIOR TO INITIATION.

Initial Operation

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* CAUTION *
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THE POWER SYSTEM STABILIZER' MUST BE OUT OF SERVICE OR THE REGULATOR IN THE MANUAL MODE FOR ALL TESTS THROUGH PARAGRAPH 9 (EXCEPT PARAGRAPH 3 AS NOTED). FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

1. Magnetic Pick-up Tests. With the turbine generator at rated speed, check the output of the magnetic pickup with an oscilloscope. This can be done by connecting the oscilloscope leads across terminals L23 and L24 or connecting the leads to "IN TEST" on the Speed Transducer board (LOC 1101). For a 3600 RPM machine and an 80 toothed wheel, this frequency will be 4800 Hz. The signal should be approximately a sine wave (some distortion is normal) and have a magnitude of at least 5 volts P-P (normally 30 to 40 volts P-P will be obtained), Amplitude modulation of the signal will exist because of shaft run-out, imperfect toothed wheels, etc. This usually appears as a once per revolution frequency (60 Hz for a 3600 RPM machine). The existence of high frequency spikes usually indicates a shield grounding problem and a check should be made for the existence of more than one ground on the shield. Pull the Speed Transducer board (LOC 1101).

2. Power Supply Tests. Turn on ±15 VDC power using the toggle switch mounted on the faceplate of the Power Supply board (LOC 1111). Check that the power supply voltages are correct. Adjustment of the ±15 volt supply may be made with board mounted rheostats. (Refer to Theory of Operation section). The current limit and the crow bar settings are factory set and need no further adjustment.

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ON SOME EXCITATION EQUIPMENT, PULLING THE RELAY BOARD INTRODUCES A STEP INTO THE REGULATOR. THIS CAN RESULT IN DAMAGE TO THE GENERATOR. INSURE THAT THE REGULATOR IS IN THE MANUAL MODE OF OPERATION BEFORE PULLING THE RELAY BOARD.

3. Relay Board Tests. If the generator is in the automatic mode of operation, transfer to manual. Pull the Relay board (LOC 1112). The Relay board is designed to allow the PSS to be used with many excitation systems. Board mounted shorting clips and rheostats are used to adapt the relay board to the particular excitation system. (Refer to the drawings furnished with the equipment for the settings of F1S, F2S, F3S, F4S, F1P and F2P.) check settings and readjust if necessary. Insert Relay board. Pull Washout and Output board (LOC 1108). Put shorting clip C5S on terminals 1, 2 (not GRD). Insert Extender board in LOC 1108).

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THE REGULATOR MUST BE IN THE MANUAL MODE OF OPERATION FOR THIS TEST. FAILURE TO PLACE THE REGULATOR IN MANUAL MAY RESULT IN DAMAGE TO THE GENERATOR.

Insert a signal from pins 11, 12, to pins 3, 4 of the Extender board. This signal can be a dc signal or an oscillating signal (peak value should be less than 10 volts). With the PSS off (as evidenced by the "OFF" LED on the Relay board being lighted) no voltage should exist at the test jack labeled "Out to Regulator" mounted on the Relay board. Pressing the "ON" pushbutton should cause the voltage to be present but attenuated for insertion into the appropriate regulator. If the "ON" LED will not stay energized, press the reset pushbutton mounted on the faceplate of the Ceiling Sensing, board. If the "ON" LED will still not stay lit, check that the permissive contact is closed (usually 52G line breaker - see sheet 1c of Figure 13). If the unit is off line, this permissive contact must be jumpered in order to establish that the output relay is working properly. Press the "OFF" pushbutton. The voltage at the test jack should be zero and the "OFF" LED should be lighted. Remove Extender board. Insert Washout and Output board in LOC 1108. Set the positive and negative limit dials on 1000 (10 turns). On Ceiling Sensing board (LOC 1110), put the delay dial on zero, the positive limit dial on 500 and the negative limit dial on 500. Press the reset pushbutton on the Ceiling Sensing board. Adjust the gain of the Washout & Output Board to 1.0 by putting the jumper C3S to 1,2 and turning the potentiometer marked "GAIN K" to zero. Connect a variable source of dc voltage to "IN 1" of the Washout and Output board and set the voltage to some value less than 5 volts. Press the "ON" pushbutton on the Relay board. A voltage should exist at the test jack on the Relay board labeled "Out to Regulator". Increase the voltage at "IN1" of the Washout and Output board to ≈6 volts. After approximately 2.5 seconds, the FAR relay should drop out causing the "OFF" LED to be lit. Try to reinsert the PSS by pressing the "ON" pushbutton. The PSS should not latch in the "on" position. Remove the voltage from IN1. Press the reset pushbutton on the Ceiling Sensing board. Put the PSS in service (with the generator control in manual) by pressing the "ON" pushbutton. The PSS should latch in the "on" position and the "ON" LED should be lit. Remove the PSS from service by pressing the "OFF" pushbutton. The "OFF" LED should be lit. Some installations may have customer controls for the "PSS" mounted in the control room. Check that any external customer controls can be used to insert and remove the PSS. Leave the PSS in the "off" position.

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THE FOLLOWING GAIN TEST MUST BE PERFORMED WITH THE PSS IN SERVICE AND THE REGULATOR IN THE AUTOMATIC MODE. INSURE THAT THE SHORTING CLIPS AND RHEOSTAT SETTINGS ON THE RELAY BOARD ARE CORRECT FOR THIS EQUIPMENT. INSURE THAT THE SPEED TRANSDUCER BOARD IS PULLED. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

The Relay board contains the necessary interface circuitry so that a one volt change in PSS output will cause a one percent change in generator terminal voltage. With the PSS off, a small step of voltage (1 VDC) should be inserted into IN1 of the Washout and Output board. The regulator should be in the automatic mode of operation with the generator preferably off-line. The test may be satisfactorily performed on-line if the reactive current compensator (if any) is bypassed or switched to zero and if the voltage regulator of any other generator units paralleled at terminal voltage are temporarily turned to manual control, Place the PSS in service. Measure the change in the generator terminal voltage. The gain is calculated by dividing the percent change in generator terminal voltage by the step in voltage applied to IN1. The gain should be (-) 1%/volt ±0.1. The minus sign (terminal voltage decreases for a positive input voltage) arises because the signal is inserted into IN1 which bypasses the washout stage. F1P (on the Relay board) can be used to adjust the gain, if necessary. Remove the voltage at IN1.

An attempt has been made to interface with the voltage regulator in such a way that placing the PSS into and out of service will not cause a bump. On some equipments, F2P on the Relay board has been furnished for this purpose. With the voltage regulator in manual, (or test if a test position is available on the control switch) pull the Speed Transducer (LOC 1101). Put the PSS in service using the pushbuttons on the Relay board. Note whether the ac regulator output voltmeter or transfer voltmeter changes when the PSS is put in service. Use F2P on the Relay board to minimize the change in ac regulator output voltage when the PSS is inserted. Remove the PSS from service.

4. Speed Transducer Tests. To check the operation of the Speed Transducer board (LOC 1101), disconnect the leads from the toothed wheel/magnetic pickup at L23, L24. Connect an audio oscillator to L23, L24. Adjust the oscillator for approximately 10 volts peak-to-peak at 4800 Hz.

   Insert the Speed Transducer board. The output of the Speed Transducer board should be zero (the dc level should be zero but there will be ≈1 volt of 4800 Hz carrier feedthrough). If the voltage is more than ± 2.5 VDC, use the null potentiometer, A1P, to get 0 volts ± 0.2 volts. Increase the frequency of the audio oscillator to 5040 Hz. The transducer output should increase by +5 volts from the 4800 Hz value (± 0.5 volts). Decrease the audio oscillator frequency to 4560 Hz and the transducer output should change by -5 volts from the 4800 Hz value (±0.5 volts). The gain should not be adjusted unless the oscillator frequency calibration is accurate and the output is off by more than 0.5 volts. Remove the audio oscillator. Reconnect the leads from the toothed wheel/magnetic pickup to terminals L23, L24. With the turbine-generator synchronized, adjust the null potentiometer, A1P, for 0 volts ± 0.2 volts at the Speed Transducer output test jack. The Speed Transducer power supplies are isolated by 12 volt Zeners. If the null cannot be adjusted to 0 volts ± 0.2 volts, remove one of the shorting clips on the P1 terminal board mounted on the back of the rack. If the minimum voltage attainable is positive, then remove the shorting clip from Pl-7 to Pl-6. If the minimum voltage is negative, remove the shorting clip from Pl-5 to P1-6.

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THE PSS MUST BE OUT OF SERVICE FOR THE FREQUENCY RESPONSE TESTS. FAILURE TO DO THIS MAY RESULT IN DAMAGE TO THE GENERATOR.

5. Frequency Response Tests. Set the Signal Conditioning board (LOC 1102), for lead time constants of 0.26 seconds and lag time constants of 0.03 seconds. This can be accomplished by the following adjustments:

TCT2V = 0 turns

TCT2F = 0.9 turns

TCT3 = 1.5 turns

TCT4 = 0.9 turns

TCT5 = 1.5 turns

On the Washout and Output board (LOC 1108) set the washout time constant for 2 seconds and the gain for 20 by setting the jumpers on C1S, C2S, and C5S to 2-3 and the jumpers on C3S and C4S to 1,2. Set the TCT1 dial on 6.8 and the gain control at 3.8. Check the Band Reject (Biquad) Filter Board (LOC 1109) to confirm that the shorting clips on 1TB, 2TB, 3TB and 4TB are on terminals 1,2. The shorting clip on 5TB should be on terminals 2, 3. Pull the Speed Transducer board (LOC 1101). Insert the Extender board in place of the Speed Transducer board. Connect the audio oscillator to terminal 15 or 16 and ground terminal 3 or 4 of the Extender board. Measure the frequency response of the PSS main channel from 0.1 Hz to 100 Hz measuring oscillator output and output test connector on the Washout and Output board.

The oscillator output level should be held at the values shown on Figure 1 to avoid saturation of the amplifiers. Record data and compare gain to values shown in Figure 1. Actual gain and nominal gain should not differ by more than 2 to 1. Figure 2 is a plot of gain (in Db) vs frequency at these settings.

6. Ceiling Sensing Tests. To check out the Ceiling Sensing board (LOC 1110), pull the Ceiling Sensing board and verify that the shorting clip on E1S is between 1 and 2. Reinsert the board. Set both the positive limit (E2P) and negative limit (E3P) for 500 (5.0 turns). Set the time delay on the Ceiling Sensing board for minimum by putting E1P at zero. Insert a variable dc voltage into IN1 of the Washout and Output board. Place the regulator in the manual mode of operation.

FREQ HZ	V(IN) VOLTS (ZERO TO PEAR)	V(OUT) VOLTS (ZERO TO PEAK)	GAIN	NOMINAL GAIN (1)
0.1	0.2			14.1
0.2	0.2			18.2
0.4	0.2			24.8
1.0	0.2			58.9
2.0	0.02			175.8
3.0	0.02			295.1
5.0	0.02			467.7
7.0	0.02			446.7
10	0.02			1257.0
12	0.02			1113.5
15	0.2			11.2
17	0.2			7.7
20	0.2			3.7
25	0.2			2.3
30	0.2			1.6
35	0.2			2.0
40	0.2			3.3
45	0.2			5.2
50	0.2			4.0
55	0.2			1.9
65	0.2			1.9
70	0.2			2.6
80	0.2			410
90	0.2			5.6
100	0.2			7.5

T_1 = 2 SEC. $T_2 = T_4$ = 0.26 SEC
T3 = T_5 = 0.03 SEC.
GAIN = 20

Gain is measured from the output of the Speed Transducer to the output of the Washout and Output board. Use an oscilloscope to ascertain that nothing is saturating.

Figure 1
Frequency Response of Main Channel

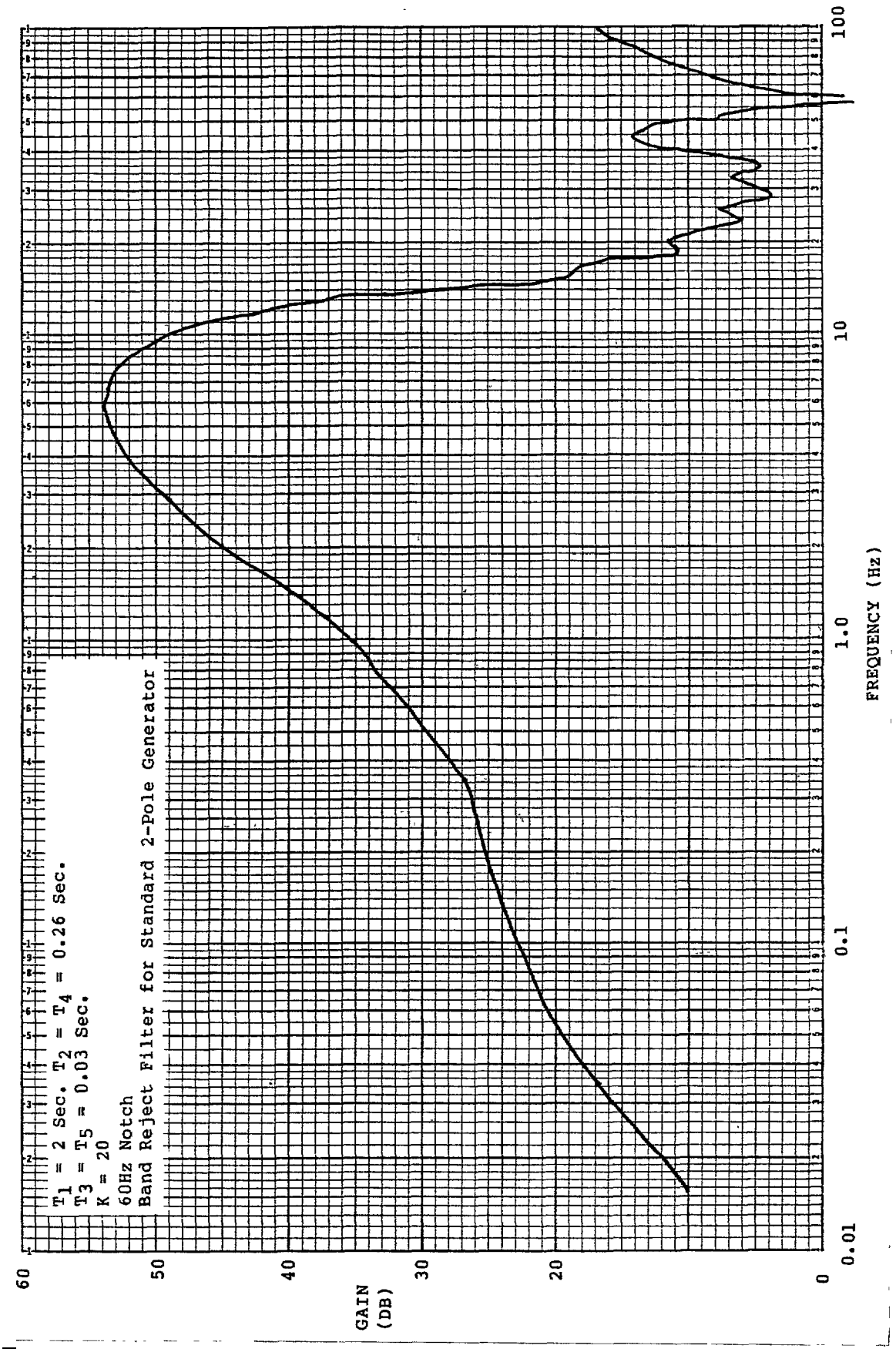


Figure 2
PSS Gain vs Frequency

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THE REGULATOR MUST BE IN THE MANUAL MODE OF OPERATION FOR THIS TEST. FAILURE TO DO THIS COULD RESULT IN DAMAGE TO THE GENERATOR.

Place the PSS in service by pressing the "ON" pushbutton on the Relay board. Slowly increase the voltage until the PSS trips out of service as evidenced by the "OFF" LED being lit. The voltage at Out 1 on the Washout and Output board should be 5 volts ± 0.25 volts. Remove the voltage. The PSS will not stay in service when the "ON" button on the Relay board is pressed and released. Press the reset pushbutton on the Ceiling Sensing board. Pressing the "ON" pushbutton on the Relay board should have no effect. Press the reset pushbutton on the Ceiling Sensing board. Pressing the "ON" pushbutton on the Relay board should now cause the PSS to be placed in service and stay on when the button is released. Reverse the voltage into IN1 of the Washout and Output board. The same procedure can be followed to verify that this part of the circuit is working. Adjust the time delay (E1P) on the Ceiling Sensing board for 300 (3 turns). Remove the voltage at IN1 of the Washout and Output board and press the reset pushbutton on the Ceiling Sensing board. Place the PSS in service (regulator in manual). Put a voltage into IN1 of sufficient magnitude to cause a trip. The PSS should trip to "OFF" approximately 10 seconds after insertion of the signal. Remove the test voltage and reset the Ceiling Sensing board by pressing the reset pushbutton on the faceplate. Place the positive and negative limit dials on 1000 (10 turns) and the time delay dial on zero. Final settings of time delay and gain can only be made after on-line tests and/or system studies.

7. Washout and Output Tests. On Washout and Output board, set negative limit dial and positive limit dial on 500 (5 turns). Insert a variable dc voltage into IN1. Monitor the output voltage at OUT1 using an oscilloscope or voltmeter. Increase the dc voltage slowly and note at what voltage the output fails to follow the input voltage (output voltage is limited). This voltage should be 4.75 volts ±0.5 volts. Reverse the voltage and perform the same test. The output voltage should be -4.75 ± 0.5 volts. Remove the voltage source and Extender board. Reinsert the Speed Transducer board. Insure that the PSS is off. Place the regulator in the automatic mode of operation. Place the positive and negative limit dials on 1000 (10 turns) and the time constant dial on zero. Final settings of time constant and gain can only be made after on-line tests and/or system studies.

8. Maximum Gain Noise Tests. The adjustment range provided by the PSS can result in extremely high gain over a range of frequencies. For instance, with K set at 100, T2 and T4 set at maximum, and T3 and T5 set at minimum, the maximum gain will be 120 Db (1,000,000:1). Although this gain will probably never be used in an actual system, a good check of PSS noise level can be obtained using this gain. Insure that the PSS is off. On the Signal Conditioning board (LOC 11021, set TCT2F and TCT4 for 1000 (10 turns) and TCT3 and TCT5 for zero.

Pull the Washout and Output board (LOC 1108). Put the shorting clip on C5S on terminals 2, 3 (GRD). Put the shorting clip on C3S on terminals 2, 3. Insert Washout and Output board. Turn gain adjustment to 1000 (10 turns). Remove Speed Transducer board (LOC 1101). Insert the Extender board in place of the Speed Transducer. Short from pin 15 to pin 3 of the Extender board. Using an oscilloscope with a very slow sweep speed (1 sec/cm), determine the peak-to-peak output voltage (measured at OUT 1 of the Washout and Output board). This voltage should be observed for some time, perhaps up to a minute. This voltage should be less than 3 volts peak-to-peak. Remove the Extender board. Insert the Speed Transducer board in LOC 1101. Remove Washout and Output board. Place C5S shorting clip on terminals 1, 2; (not GRD). Place C3S shorting clip on terminals 1, 2. Return gain adjustment dial to zero (unity gain). Insert Washout and Output board. Final settings of time constants and gain can only be made after on-line tests and/or system studies.

Initial Settings

The PSS has been designed to interface with various excitation systems. In order to permit compensation for widely different systems, adjustments are made using calibrated dials. On the Washout and Output board, set the gain dial to zero and C3S on terminals 1 and 2. Insure that the Relay board has the correct rheostat and switch settings. Check the Speed Transducer board to insure that A10C is not in the circuit. Check that B1S on the Signal Conditioning board has the shorting link on 1, 2 (GND).

Settings for a particular installation are best obtained by analytical studies and/or on-line tests. It is the responsibility of the customer to perform the analytical studies and on-line tests necessary to determine proper settings. General Electric Company can provide assistance in performing analytical studies and system tests. Contact your local sales office. A paper, "Concepts of Synchronous Machine Stability as Affected by Excitation Control", by F. P. deMello and C. Concordia may be used as a reference (68-TP-129-PWR). Other publications which may be used as references are "Test Procedure for Power System Stabilizers" published by the Western Systems Coordinating Council and an IEEE publication "Guide for the Identification, Testing and Evaluation of the Dynamic Performance of Excitation Control Systems" paragraph 6.03 (P421, 1977).

On-Line Tests

Power System Stabilizers are employed to provide an additional input to the regulator to improve power system dynamic performance. They have been employed to solve two separate but related problems: local machine-system oscillations in the l-2 Hz range and inter-area oscillations at 0.5 Hz or less. The fundamental function of the signal conditioning network of the PSS is to compensate for the phase lags of the system being controlled. For best contribution to system damping, the gain of the PSS should be maximized under the most sensitive conditions (usually at or near full-load). The desired stabilizing signal has as its reference the base system frequency which is slowly changing all the time. The washout function minimizes the effect of overall system speed changes on machine terminal voltage.

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THE PSS HAS A WIDE RANGE OF ADJUSTMENT OF LEAD AND LAG TIME CONSTANTS. DO NOT ADJUST FOR A DIFFERENCE IN LEAD-LAG SETTINGS OF MORE THAN TEN TO ONE. LARGER DIFFERENCES IN SETTINGS CAN CAUSE EXCESSIVE AMPLIFICATION OF TORSIONAL SIGNALS AND RESULT IN DAMAGE TO THE TURBINE-GENERATOR.

The PSS has adjustable limits on the output signal so that the PSS will not completely override voltage regulator action. The output voltage limits are independently adjustable but should be set identically in order to prevent nonsymmetrical limiting which might bias the voltage regulator. Since the ceiling sensing function occurs after the adjustable output limit stage, the output voltage limits should be set at a higher voltage than the ceiling sensing limits.

In order to facilitate on-line frequency response testing of the excitation system, IN1 is provided on the Washout and Output board. With the shorting clip on C5S on terminals 1, 2, a drive signal can be inserted into IN1 which will produce a calibrated terminal voltage change through the regulator. For normal operation, the shorting clip on C5S should be on terminals 2, 3 (GRD) and the shorting clip on C1S should be on terminals 2, 3.

Figure 3 is provided so that dial settings can be chosen to realize the desired time constants and limit settings.

PRINCIPLES OF OPERATION

General

The Power System Stabilizer (PSS) is used to provide an additional input to the voltage regulator which, under certain conditions, can improve power system dynamic stability. A signal is obtained from a toothed wheel/magnetic pickup mounted at the turbine front standard. This signal is frequency modulated with the carrier frequency being 4800 Hz (for a 3600 RPM machine and 80 toothed wheel) and the modulation frequency being equal to the speed changes of the turbine shaft. This frequency modulated signal is passed through a speed transducer to produce an output signal proportional to the magnitude of the deviation in speed with a frequency equal to the speed deviation frequency. This demodulated signal is then passed through two stages of lead-lag compensation to provide phase compensation for the system, a twin-tee notch filter tuned to 60 Hz (30 Hz for an 1800 RPM machine), a band reject filter to attenuate signals at torsional frequencies, a washout stage to remove drift and slow system frequency changes and an adjustable gain stage to maximize the effect of the PSS. Limits are provided on the output stage so that during large system disturbances, the PSS will not interfere with voltage regulator action. A relay is provided for placing the PSS in service (and also removal from service). Interfacing is provided to properly scale the PSS output signal so that 1 volt from the output stage produces a 1% change in terminal voltage. A protective circuit is provided to detect a failure which causes the output to go to ceiling (Ceiling Sensing board). Ten-turn rheostats with calibrated dials are furnished for all customer adjustments. Figure 13 contains an overall block diagram and a functional elementary of the PSS. Figures 4-7, 9-11 are the elementary diagrams of the printed circuit boards.

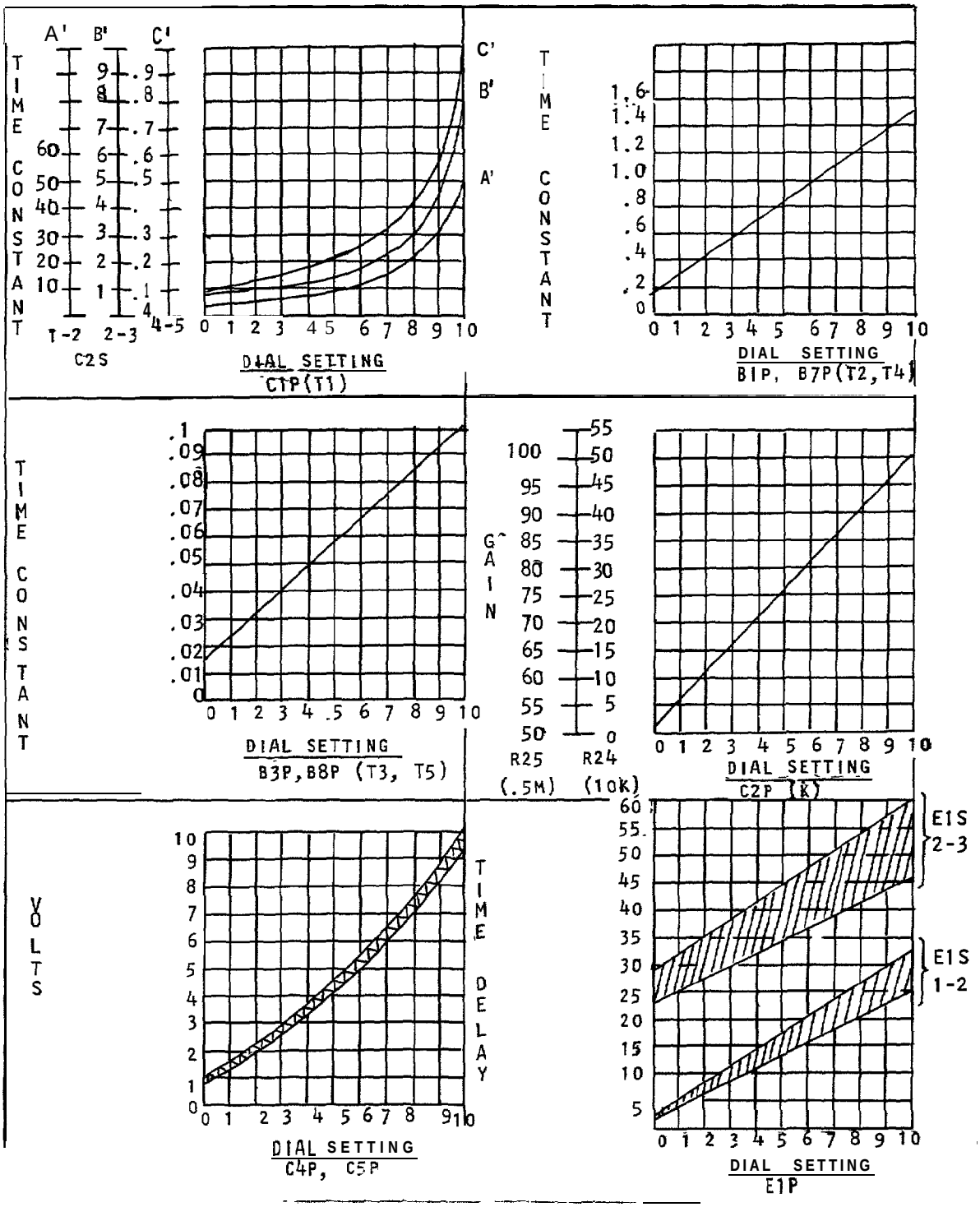


Figure 3
Parameter Value vs Dial Setting

Power Supply System (Figure 13, Sheets 1C and 1D, & Figure 4).

The panel-mounted PSS includes both the ± 15 VDC and 24 VDC power supplies. The 115 VAC, 60 Hz power should be connected to L21 and L22. S1T and S2T transform the input voltage and provide isolation. The voltage is rectified by a single phase full-wave bridge, filtered to remove the AC component, limited by Zener diode action and then fed to series pass regulators. Fuses R1F and R2F, SCRs R1CR and R2CR, along with associated sensing and firing circuits form a "crowbar" circuit which will protect sensitive semiconductor components from overvoltage by blowing a fuse and thus removing the excess voltage. IC voltage regulators (G1A, G2A) are used to control the output voltage. G2P and G4P, mounted on the printed circuit board, provide adjustment of the output voltage. G1R, G1P, G2R, and G3R provide a current limit feature on the +15 volt regulator (a similar circuit is found on the -15 volt regulator). The current limit is of the foldback type, i.e., the maximum output current is reduced as the load on the regulator is increased above the full-load value. This insures that critical components in the regulator (series pass transistors) will not be damaged by a short circuit. The current limit can be adjusted using G1P and G5P. The positive limit is factory set at 1 amp and the negative limit at 750 Ma (these are not the short circuit current values). Integrated circuits J1A and J2A provide voltage sensing for the "crowbar" circuit. J1P and J2P are factory adjusted to cause the "crowbar" circuit to operate at a supply voltage of 17 volts. S1T and S2T have taps which will permit variations in the input voltage from the nominal 115 VAC value. (Refer to the diagram for the correct tap for a given input voltage.) In order to reduce the noise level to as low a value as possible, the ± 15 volt power supply is very tightly regulated and heavily filtered. The regulation is 0.1% and the ripple on the power supply should be ≤ 5 Mv P-P. The 24 volt supply does not require tight regulation since it is used to drive relays and energize lights. S3T is used to isolate the 24 volt supply from the ±15 volt supply and to transform the input voltage to a lower voltage. This voltage is rectified by a full-wave bridge and then filtered. Regulation is provided by a Zener diode (S1Z). The 24 volt power supply is ungrounded while the +15 volt supplies are grounded through a parallel resistor capacitor combination. The high impedance ground on the ±15 volt supplies allows the PSS to be used with both grounded and ungrounded systems without ground loop problems.

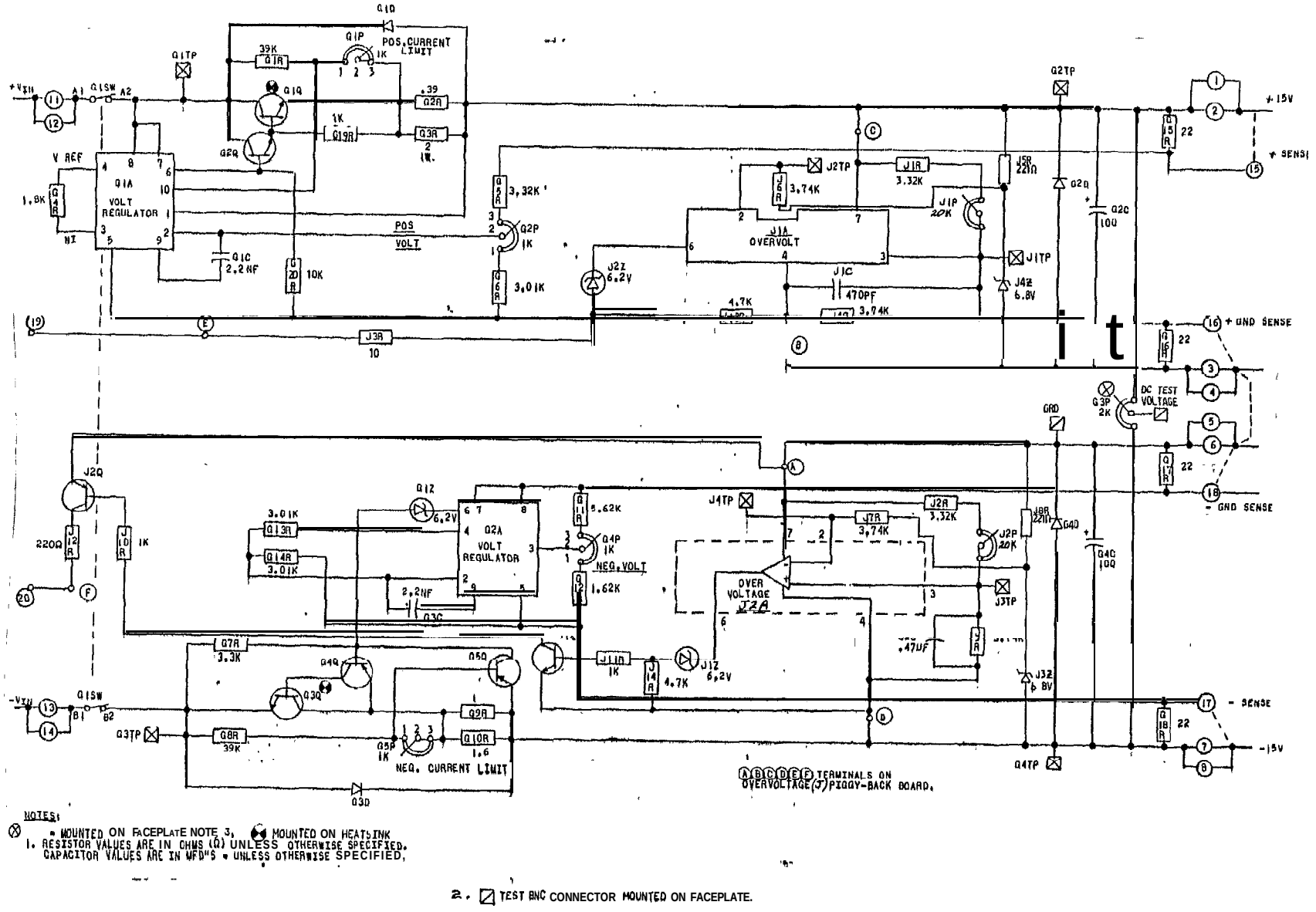


Figure 4. Power Supply

Speed Transducer Board (Figure 5)

The input signal is transformer coupled through 1T. The 1Q is used as a zero crossing detector and with 2Q forms a high gain stage so that whenever a zero crossing is detected, the voltage at 4TP will increase abruptly to 15 volts. This sudden transition in voltage is capacitively coupled through 1C to a monostable multivibrator composed of 3Q, 4Q and associated circuitry. A pulse is generated at 6TP that has an off-time determined by 2C and 8R. During the time that the input signal is negative, 1D protects 1Q and the monostable multivibrator is not triggered. 6TP stays at -5 volts. When the input voltage goes positive by approximately 1 diode drop (VBE), a pulse is generated which triggers the monostable multivibrator and the voltage at 6TP goes to approximately -15 volts. This pulse lasts for a period of time determined by 2C and 8R. The average value of the signal at 6TP is thus determined by the ratio of the on time to off time (on is -5 volts: off is -15 volts). The off time is fixed by 2C and 8R, therefore, the average voltage at 6TP will vary with input frequency. The signal at 6TP is fed through a low pass filter to summing amplifier 1A. A signal is taken from 1P to null the output of 1A at rated frequency so that deviations in frequency are detected. The 2P and 17R are used as feedback for 1A. The 2P is adjusted to give a 1 volt change for a 1% change in input frequency. Provisions have been made so that the feedback time constant can be selected for 280 Hz or 5 Hz. It should be set for 280 Hz on units with a band reject filter by disconnecting A10C from the circuit by means of the jumper.

Signal Conditioning Board (Figure 6)

The Signal Conditioning board consists of two lead-lag stages and a twin-tee notch filter. In addition to the signal from the Speed Transducer, an extra input with separate lead time constant adjustment is provided for a voltage input (IN2). B1P provides adjustment of the first stage lead time constant, which is proportional to B1C, B1R and the setting of B1P. B3P provides adjustment of the lag time constant which is proportional to B9R, B4C and the setting of B3P. The dc gain of the stage is B9R/B1R, which is 1.0. B3R, B3C, and the equivalent resistance of B3P cause extraneous lead and lag time constants which have been minimized but may be significant when performing calculations at torsional frequencies. The second lead-lag stage is identical to the first except for the addition of an output voltage limit circuit formed by B1Z, B2Z, B5D, B6D and a drive circuit formed by B1Q, B21R, B22R, B3D and B4D to increase the output current capability of B2A. A twin-tee notch filter is placed between the first and second lead-lag stages to remove 60 Hz. The twin-tee is factory adjusted to produce greater than 50 Db attenuation at the notch frequency. The notch frequency is given by:

$$F = \frac{1}{2\pi(B11R + B4P)(B16C)}$$

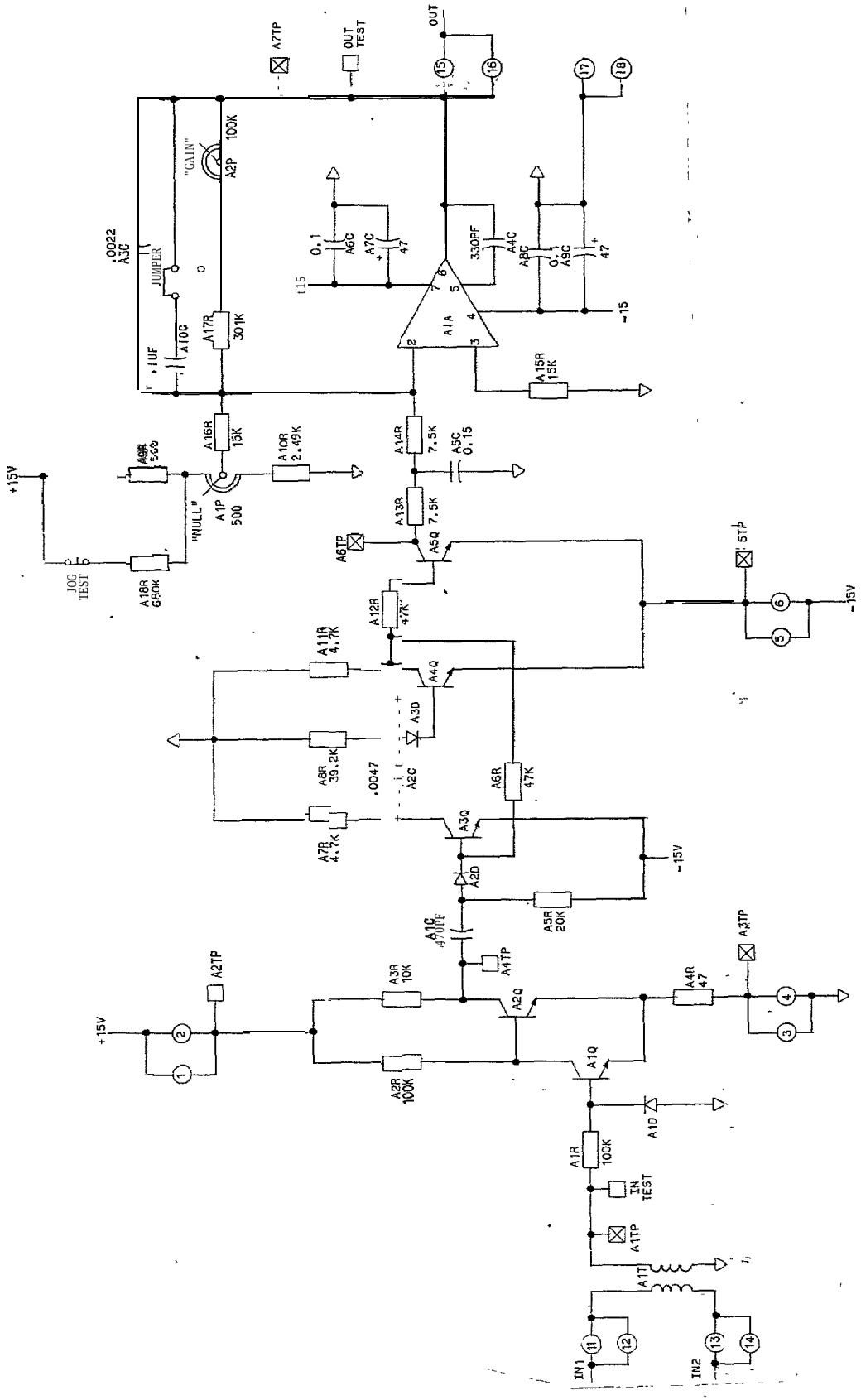
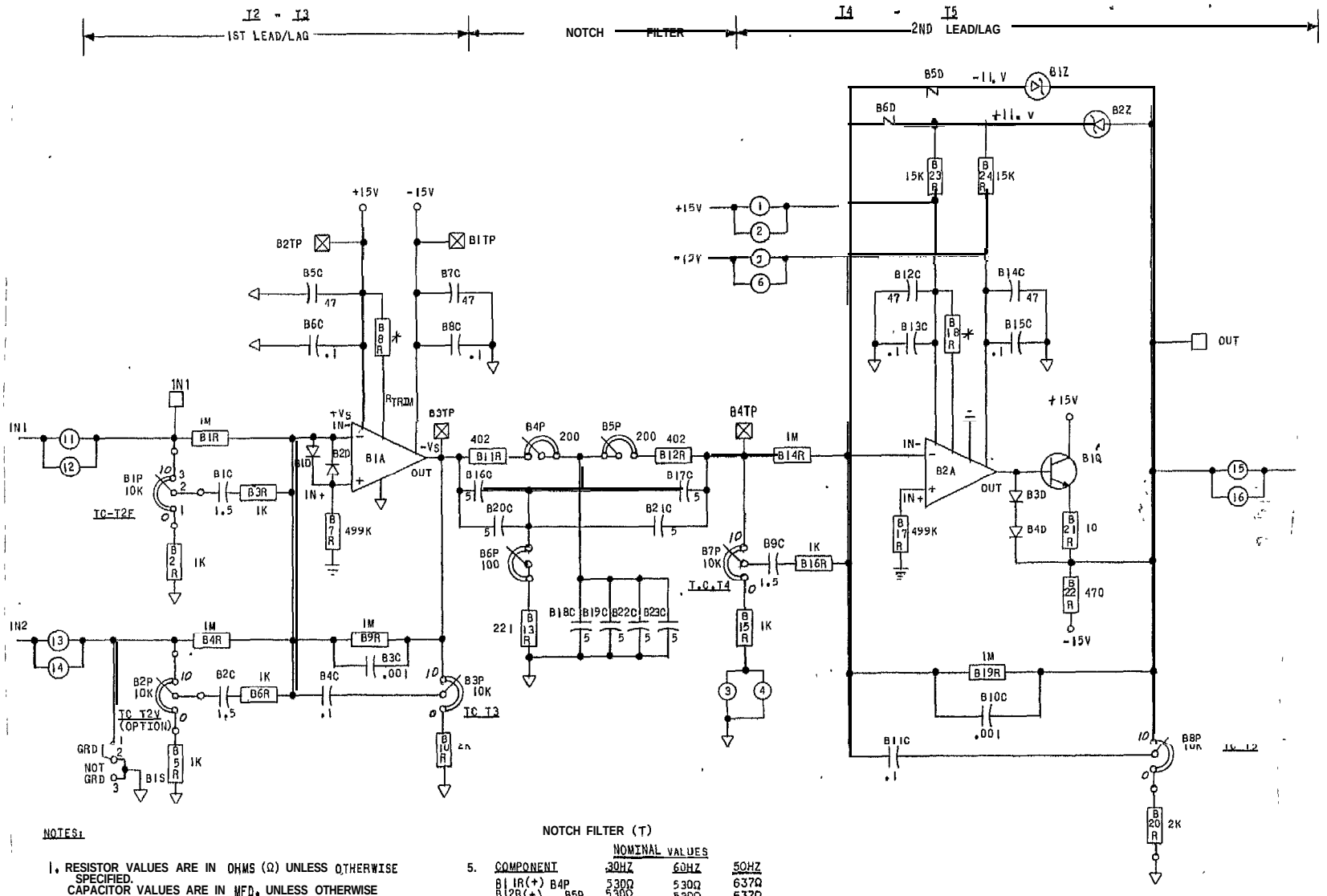


Figure 5. Speed Transducer



NOTES:

1. RESISTOR VALUES ARE IN OHMS (Ω) UNLESS OTHERWISE SPECIFIED.
   CAPACITOR VALUES ARE IN MFD, UNLESS OTHERWISE SPECIFIED.
2. OMIT B20C, B21C, B22C, B23C ON 3600 RPM GENERATORS (60HZ).
3. * RESISTOR VALUE WILL CHANGE FROM ONE UNIT TO NEXT.
4. ❑ BOARD MOUNTED TEST POINT,
6. ☐ TEST BNC CONNECTORS ON FACEPLATE OF BOARD.

NOTCH FILTER (T)

5.

COMPONENT	NOMINAL VALUES 50HZ	60HZ	50HZ
B1 1R(+) B4P	530Ω	530Ω	637Ω
B12R(+) B5P	530Ω	530Ω	637Ω
B13R(+) B6P	265Ω	265Ω	318Ω
B16C	10MFD	5MFD	5MFD
B17C	10MED	5MFD	5MFD
B18C, B19C B22C, B23C	20MFD	10MFD	10MFD

Figure 6. Signal Conditioning Board .

Band Reject (Biquad) Filter (Figure 7)

The Band Reject (Biquad) Filter is a five-stage filter used to attenuate signals at the torsional resonant frequencies of the turbine generator. It provides unity gain outside the rejection band and approximately 40 Db attenuation in the rejection band. Figure 8 is a calculated attenuation versus frequency curve. In each stage, amplifiers 11A, 12A and 13A with associated components comprise a resonant circuit. The output of 12A is summed with the input signal in 14A to obtain an output signal. Signals not within the pass band of the resonant circuit are passed with no attenuation. Signals that fall within the pass band of the resonant circuit are passed and then subtracted from the original signal to give a reject characteristic. The transfer function for each stage is described by the following equation:

$$\frac{E_o}{E_i} = \frac{1 + \frac{\alpha_N}{\omega_o} S + \left(\frac{S}{\omega_o}\right)^2}{1 + \frac{\alpha_D}{\omega_o} S + \left(\frac{S}{\omega_o}\right)^2}$$

Where α_N, α_D, ω_o and α_N/α_D are determined by the value of the resistors and capacitors used. (See sheet 2 of Figure 7.)

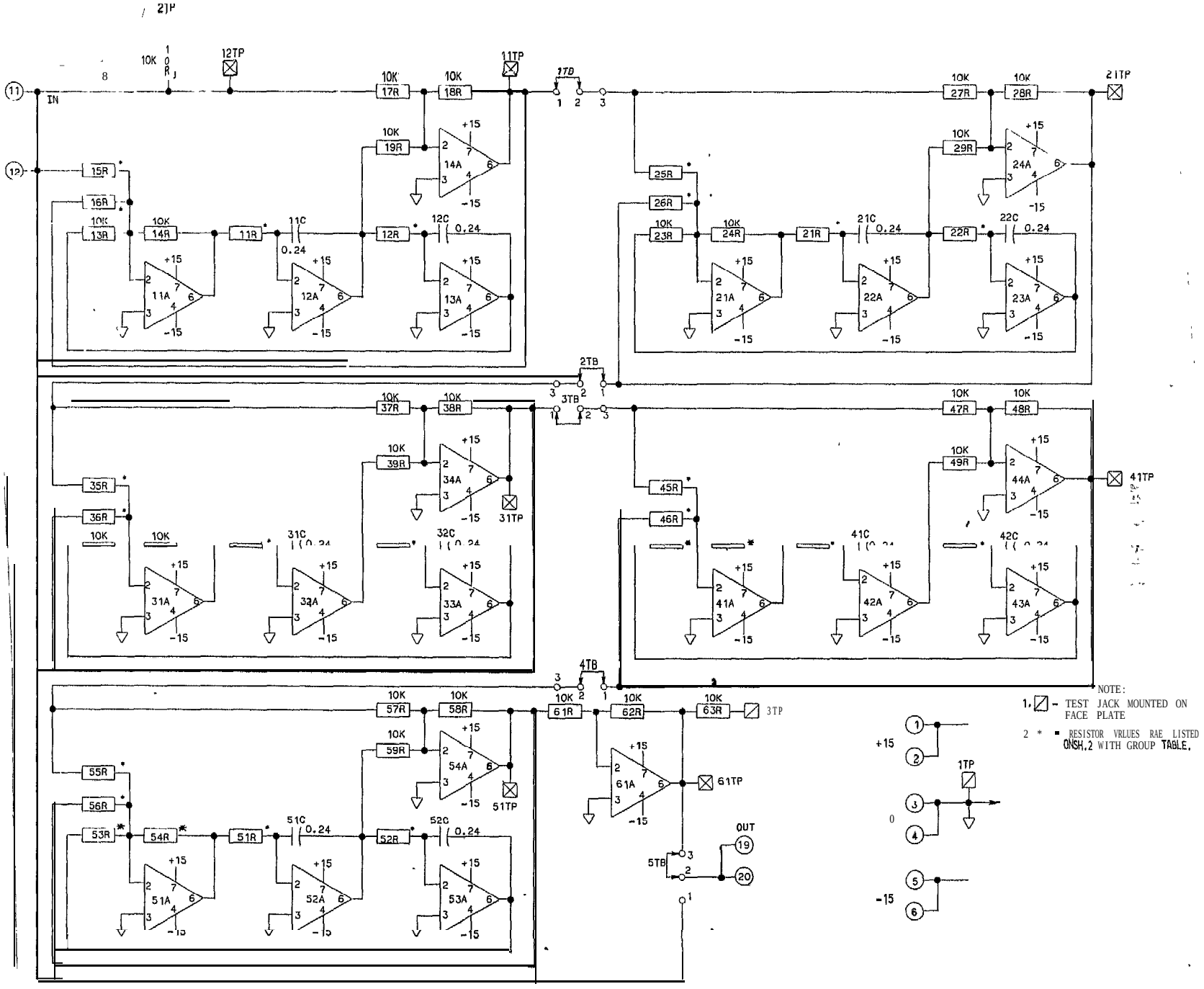


Figure 7, Sheet 1. Five Stage Cascade Band-Reject Filter

	TYPE	G01 B.R.	G02 B.P.	G03 B.R.	G04 B.P.	Go5 B.R.	Gob B.P.
	11R	43.6'	43.6K	94.7K	94.7	55.04K	55.04K
	12R	43.b	43.6K	94.7K	94.7	55.04K	55.04K
	15R	83.3	8.33K	100.0K	10.0	125.0K	16.67K
	16R	8.33	83.3K	10.0K	100.0	16.67K	125.0K
	21R	35.01	35.0K	82.8K	82.8	51.41K	51.41K
	22R	35.0'	35.0K	82.8K	82.8	51.41K	51.41K
	25R	03.3	8.33K	50.0K	16.7	125.0K	12.50K
	26R	a.33	83.3K	16.7K	50.0	12.50K	125.0K
	31R	28.1	28.1K	73.61	73.6	48.05K	48.05K
	32R	28.1'	28.1K	73.6K	73 .b	48.05K	48.05K
	35R	83.3	8.33K	100.0K	10.0	125.0K	16.67K
	36R	a.33	83.3K	10.0K	100.0	16.67K	125.0K
	41R	22.5	22.5K	44.2K	44.2	21.59K	21.59K
*	42R	22.5	22.5K	44.21	44.2	31.09K	31.09K
	45R	83.3	8.33K	50.0K	8.33	100.0K	8.333K
	46R	8.33	83.3K	8.33K	50.0	8.333K	100.0K
	51R	18.0	18.0K	31.61	31.6	19.18K	19.18K
**	52R	18.0	18.0K	31.61	31.6	27.62K	27.62K
	55R	83.3	8.33K	33.31	8.33	100.0K	8.333K
	56R	a.33	83.3K	8.331	33.3	8.333K	100.0K
	W01	95.6	95.6	44.0	44.0	75.7	75.7
	W02	119.0	119.0	50.3	50.3	81.05	81.05
	wo3	148.3	148.3	56.6	56.6	86.71	86.71
	wo4	185.2	185.2	94.3	94.3	160.8	160.8
	W05	231.5	231.5	131.9	131.9	181	181
	α D1	1.20	n.12	1.00	0.10	O-6	0.03
	α D2	1.20	3.12	0.60	a.20	0.8	0.08
	α D3	I.20	0.12	1.00	0.10	0.6	0.08
	α D4	I.20	0.12	1.20	a.20	1.2	0.1
	α D5	1.20	3.12	1.20	a.30	1.2	0.1
	α N1	a.12	I.20	a.10	1.00	0.08	0.6
	α N2	a.12	I.20	a.20	*0.60*	0.08	0.8
	d N 3	0.12	1.20	0.10	1.00	0.08	0.6
	α N4	0.12	.20	0.20	1.20	0.1	1.2
	α N5	0.12	.20	0.30	1.20	0.1	1.2
	$\frac{\alpha N1}{\alpha D1}$	1/10	10	1/10	10	2/15	15/2
	$\frac{\alpha N2}{\alpha D2}$	1/10.	10	t/3	3	1/10	10
	$\frac{\alpha N3}{\alpha D3}$	1/10	10	1/10	10	2/15	15/2
	$\frac{\alpha N4}{\alpha D4}$	1/10	10	1/6	6	1/12	12
	$\frac{\alpha N5}{\alpha D5}$	1/10	10	/4	4	1/12	12
*	43R	10K	10K	10K	10K	8.333K	8.333K
	44R	10K	10K	10K	10K	8.333K	8.333K
**	53R	10K	10K	10K	10K	8.333K	8.333K
	54R	10K	10K	10K	10K	8.333K	8.333K

Figure 7, Sheet 2. Five Stage Cascade Band-Reject Filter

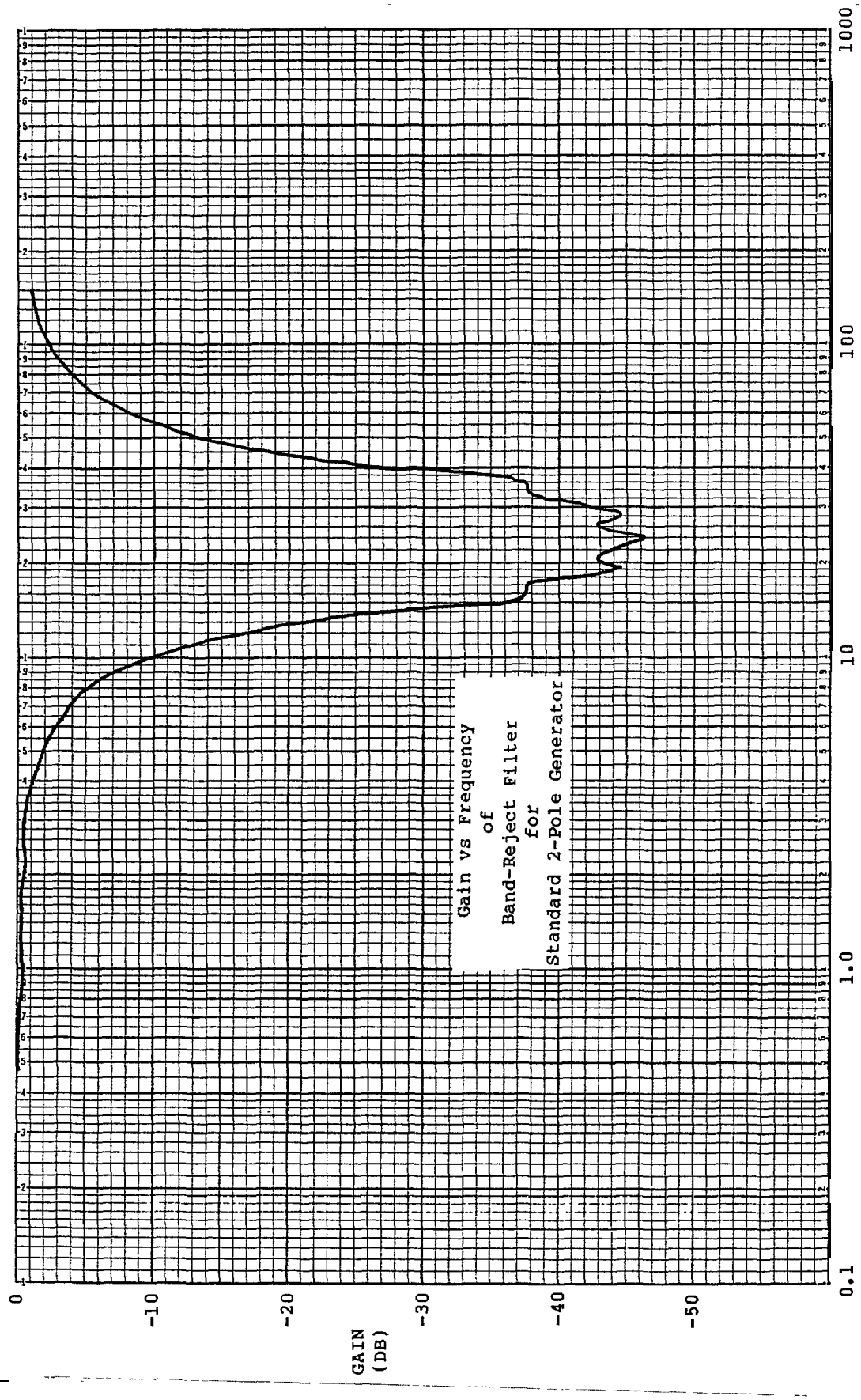


Figure 8
Band-Reject Filter for 2-Pole Generator

Washout and Output Board (Figure 9)

The Washout and Output board contains a washout function, a gain function and a power amplification function. Operational amplifier C1A is a chopper stabilized device for low offset. C1A is basically a derivative circuit formed by C1C and the feedback resistor (C2R, C3R or C4R). C2C is added to make the derivative circuit only effective for low frequencies and C1P is used to provide adjustment of the effective feedback resistance and thus control the time constant. Coarse adjustment of the time constant is available through C2S. C9D, C10D, Z1C, Z2C, C31R and C32R form a clamp to limit the Output voltage of the washout stage. An extra input is available at IN1 which bypasses the washout stage. This allows insertion of a signal into the voltage regulator for calibration of the PSS or for frequency response testing of the system. C5S is used to ground IN1 during normal use. Changing C5S shorting clip to terminals 1, 2 allows IN1 to be used for driving the regulator. C2A is used as an adjustable gain stage with the gain being determined by C3S and C2P. C3S is used to select a range of gain from 1 to 50 or 50 to 100. An adjustable limit is provided on the output of C2A by C1Q, C2Q, C15R, C16R, C1D, C2D, C4P, C5P, C13R and C14R. The output of C2A is used as the input signal for Ceiling Sensing board through pins 13, 14. C3Q and C4Q are junction FETs used as a static switch: Input IN3 is not used on this equipment. Therefore, C1S should have the shorting clip between terminals 2, 3 (OUT). C3Q will be conducting and C4Q will be non-conducting. C3A is connected to a push-up transistor stage (C7Q, C8Q, C9Q, C10Q) in order to increase the output current capability to ± 100 Ma. C3A and the push-pull stage are connected as a unity gain voltage follower.

Relay Board (Figure 10)

The Relay board is used to control insertion of the PSS into the regulator, provide the proper interfacing with different voltage regulators and provide interfacing with customer equipment. Meter M1 is mounted on the faceplate and can be used to insure that the PSS output is zero before insertion into the voltage regulator. "OFF" and "ON" pushbuttons are mounted on the faceplate and can be used to insert or remove the PSS from service. LEDs mounted near the pushbuttons indicate the status of the PSS. A line breaker contact (52G) should be used as a permissive contact so that the PSS is inoperative off-line. The pushbuttons have momentary contacts and so F93 is sealed in by an FAR and F93 contact. The F93A is an auxiliary relay used to increase the number of contacts of F93. FAR is controlled by the Ceiling Sensing board.

The F1R, F2R, and F1P are used to properly scale the PSS signal for insertion into the regulator. The F2P is used to dummy load the regulator so that a bump in excitation will not occur whenever the PSS is placed in service. (Refer to the diagram for the proper settings of F1P and F2P and switches F1S, F2S, F3S and F4S.)

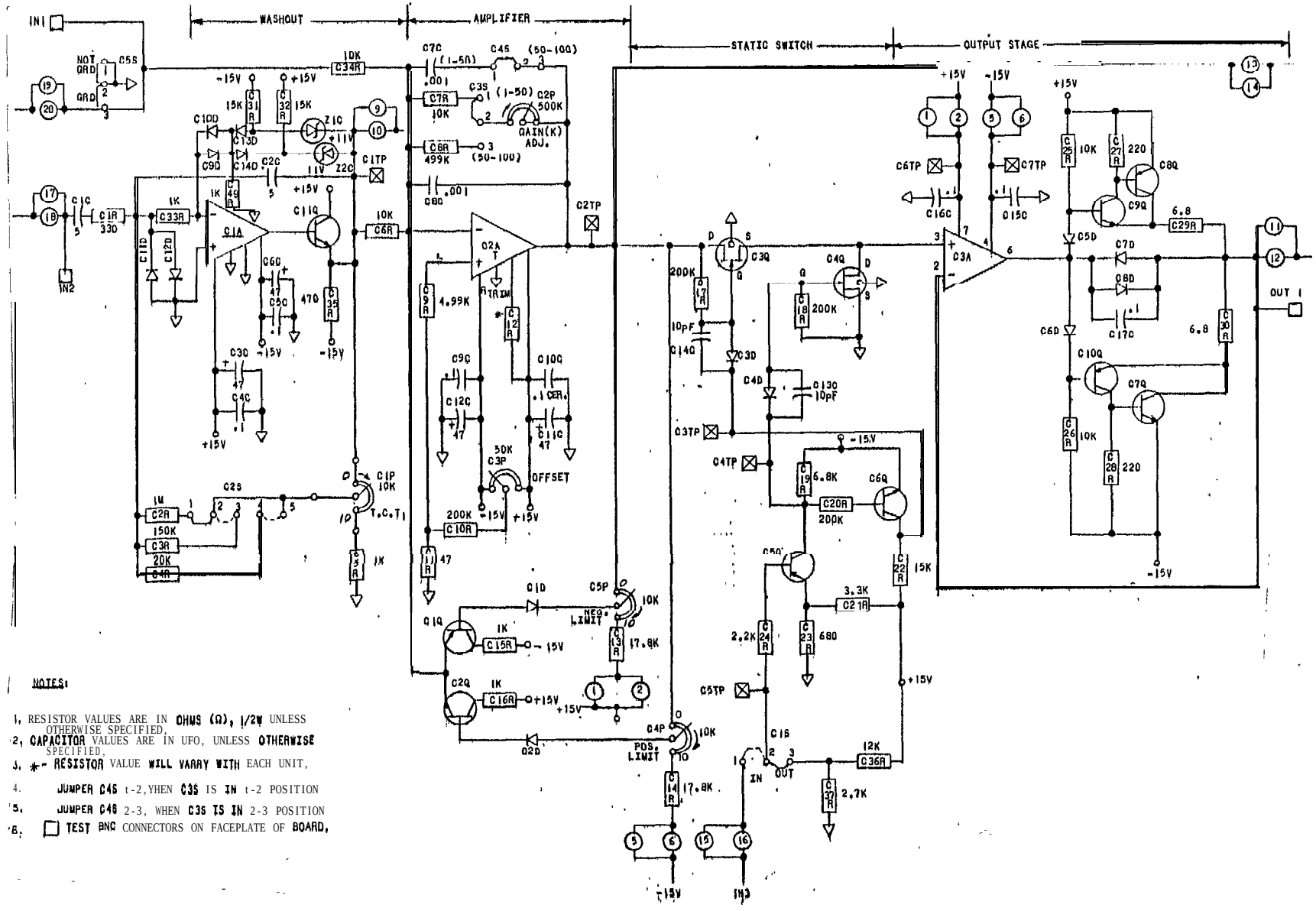
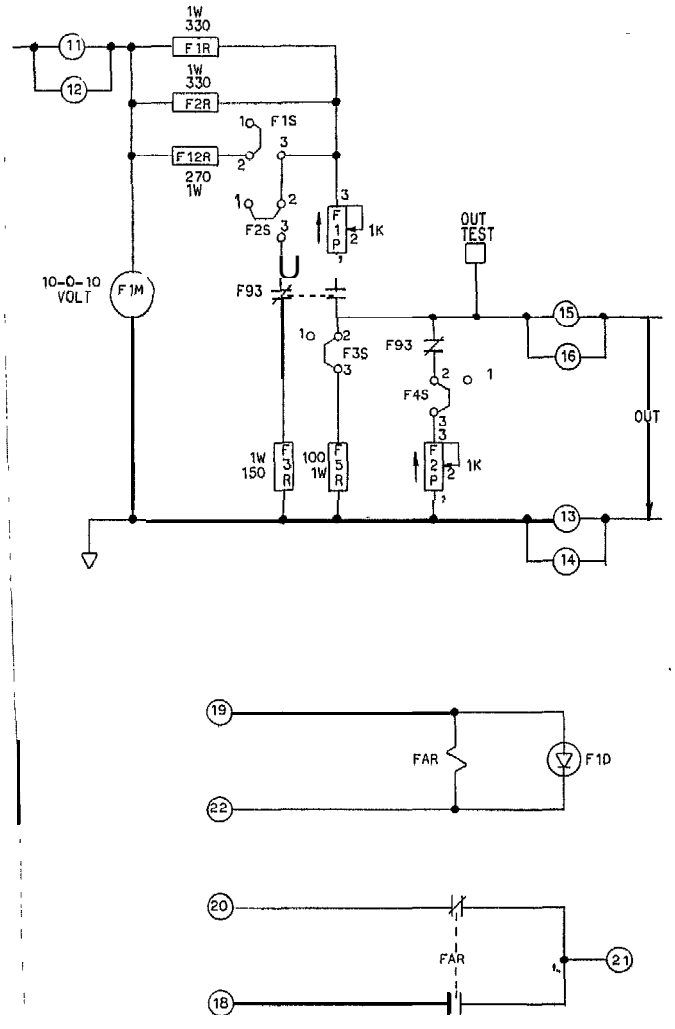
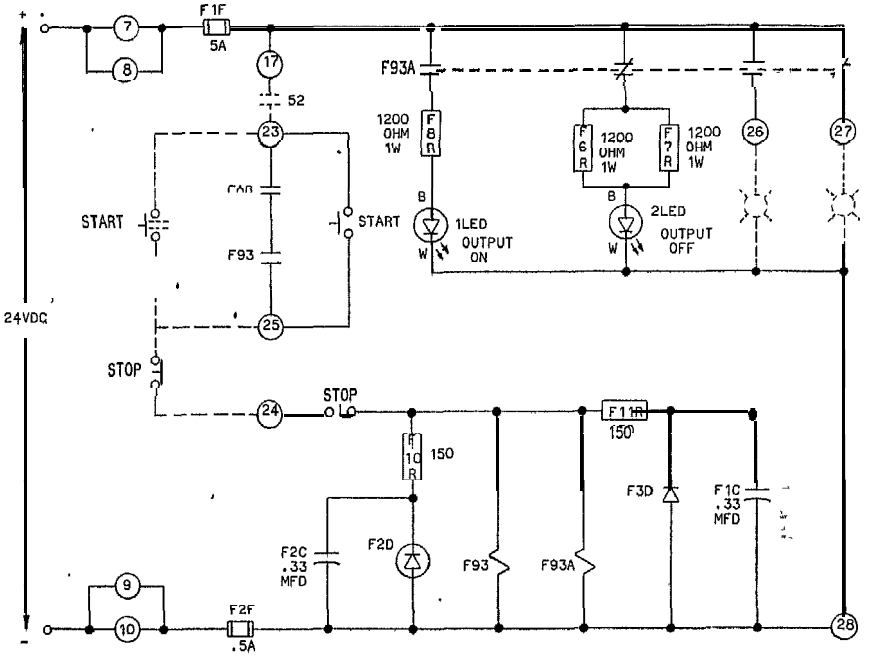


Figure 9. Washout & Output Board



NOTE. RESISTANCE IN OHMS UNLESS OTHERWISE NOTED.



JUMPER SELECTION AND POTENTIOMETER SETTINGS FOR REGULATOR TYPES.

JUMPER	GENERREX	ALTERREX	TYPE 143	STATIC	ALTHYREX
F1S	1 & 2	1 & 2	1 & 2	2 & 3	1 & 2
F2S	2 & 3	1 & 2	1 & 2	2 & 3	1 & 2
F3S	1 & 2	2 & 3	1 & 2	1 & 2	1 & 2
F4S	2 & 3	2 & 3	2 & 3	2 & 3	2 & 3
POT					
F1P	0 OHM	225 OHM	158 OHM	0 OHM	305 OHM
F2P	0 OHM	390 OHM	323 OHM	102 OHM	470 OHM

Figure 10. Relay Board

Ceiling Sensing Board (Figure 11)

The Ceiling Sensing board is used to detect a failure in the PSS which causes the output to be greater than a specified value for an adjustable length of time. Once the condition exists, the PSS cannot be restored to service until a reset pushbutton is depressed on the Ceiling Sensing board.

The E1A and E2A operate as comparators with E2P and E3P used to adjust the level at which a failure is detected. If the input voltage is less than the positive limit (E2P) and more positive than the negative limit (E3P) the voltage at E6TP is at a minimum. The voltage at E5TP is passed through an inverter in E1L and then a nand operation is performed on the signal from E6TP and the inverted signal from E5TP. A nand gate has its output in the high state unless all inputs are in the high state.

In this case, both inputs are in the high state so its output is in the low state. The output of this nand gate is passed through two inverters (no change in state) and is also used to trigger a one-shot (E3L). With a ceiling condition not detected, the one-shot trigger input will stay in the low state and the output (the complemented output is used in this case) stays in the high state. The inputs to the first nand gate of E2L are a low and a high, which means the output will be in the high state. Two of the nand gates are used in a cross-coupled fashion to allow latching whenever a ceiling condition is detected. E8C is provided to insure that the cross-coupled gates will go to the proper state when energized. With no ceiling condition detected the inverter output of E2L will be in the low state, turning on transistors E2Q and E3Q and energizing relay FAR on the Relay board.

If a ceiling condition exists (whether positive or negative), the second nand gate of E1L will have its input states changed to one low and one high. The output of this nand gate will go to the high state which triggers the one-shot (E3L) and changes the input on the first nand gate of E2L to a high state. Nothing happens however, because the output of the one-shot (E3L) is in the low state and remains there for a period of time determined by E7C, E1P and E8R or E9R (depending on the setting of E1S). After the timing interval, the output of the one-shot (E3L) goes to the high state and the output of the first nand gate of E2L goes to the low state. The output of the cross-coupled gates of E2L goes to the low state and will remain low even if the ceiling condition is cleared. The inverter output of E2L changes to the high state, which turns off E2Q and E3Q and drops out relay FAR on the Relay board, giving an alarm and tripping the PSS. To put the PSS back in service, it is necessary to press the reset pushbutton (E1PB). This changes the input to one of the cross-coupled gates from a high to a low state which forces the output of the inverter to be in a low state. The cross-coupled gates latch in this state, the relay FAR is picked up and the alarm is cleared.

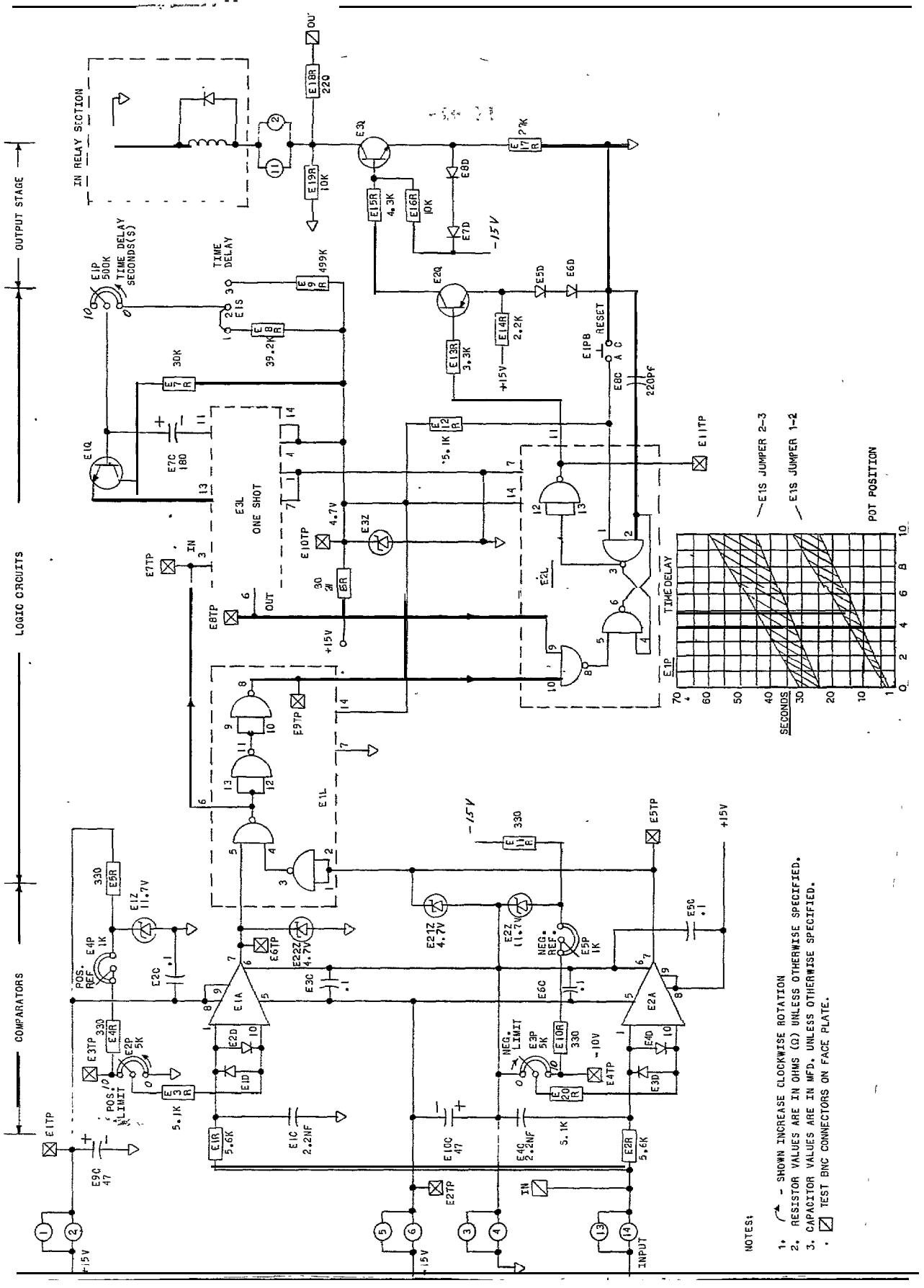


Figure 11. Ceiling Sensing board

MAINTENANCE

The PSS is static, having no moving parts except for relays and instruments. Therefore, little maintenance should be required. If excessive vibration is present, all screw connections should be checked periodically. The Speed Transducer null adjustment should be checked periodically and readjusted if necessary. A pushbutton is furnished on the Speed Transducer which puts a small step into the Speed Transducer output. This step is equivalent to approximately 0.030% change in speed. This change is so small that system noise may mask the effect on some systems. If a response can be observed in the output of the Washout and Output board when pressing the pushbutton, record for future reference. A quick check on the operating status of the PSS can then be made by pressing the pushbutton and comparing the output with the recorded output. Figure 12 shows the output of the Washout and Output board (T2 = T4 = 0.5 sec, T3 = T5 = 0.05 sec, T1 = 10 sec and K = 1.0) when the pushbutton is pressed and then released.

TROUBLESHOOTING

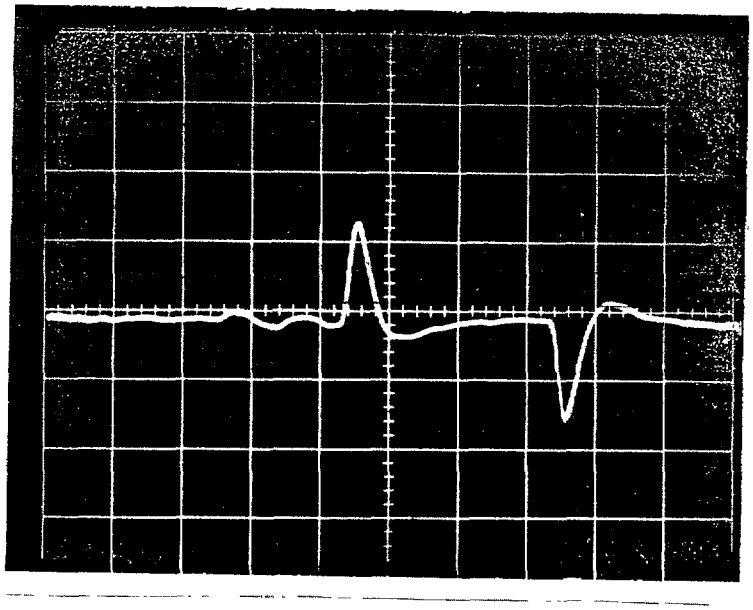
1. Carefully check all wiring if problems are experienced at start-up.
2. If no response is obtained while making the frequency response test:
   a. Check that the power switch on the Power Supply board is on.
   b. Check fuses S1F and S2F on the Power Supply panel.
   c. Check that all boards are fully inserted.
   d. Check for proper jumper connections on Washout and Output board.
   e. Check each board individually for a response.
3. If an incorrect signal is obtained in terminal voltage, check the switch and rheostat settings on the Relay board.
4. If the PSS will not stay in service:
   a. Reset Ceiling Sensing
   b. Check that permissive contact is closed.
5. Speed Transducer output stays at ceiling:
   a. Insure that generator speed is at rated.
   b. Readjust null.
   c. Remove appropriate shorting clip on P1 terminal board (see Speed Transducer Initial Operation section).

6. An instability is observed with gain dial at zero:

   a. Measure gain to ensure that it is unity.

   b. Insure that an increase in frequency causes an increase in generator field voltage.

7. If a board is found to be defective, replace it and return the defective board for repair and return.



PSS JOG TEST

Vert. = 500 mv/div
Horiz. = 200 msec/div
T1 = 10 sec
T2 = T4 = 0.5 sec
T3 = T5 = 0.05 sec
K = 1.0

Figure 12

RENEWAL PARTS

When ordering renewal parts, the following information should be given:

1. Catalog number stamped on the part, with complete description including use and location. Board components should not be replaced: order the entire board.

2. Complete nameplate data appearing in the assembly that included the component, including the serial number of the assembly.

3. If possible, data on original order on which equipment was first supplied, including all numerical references.

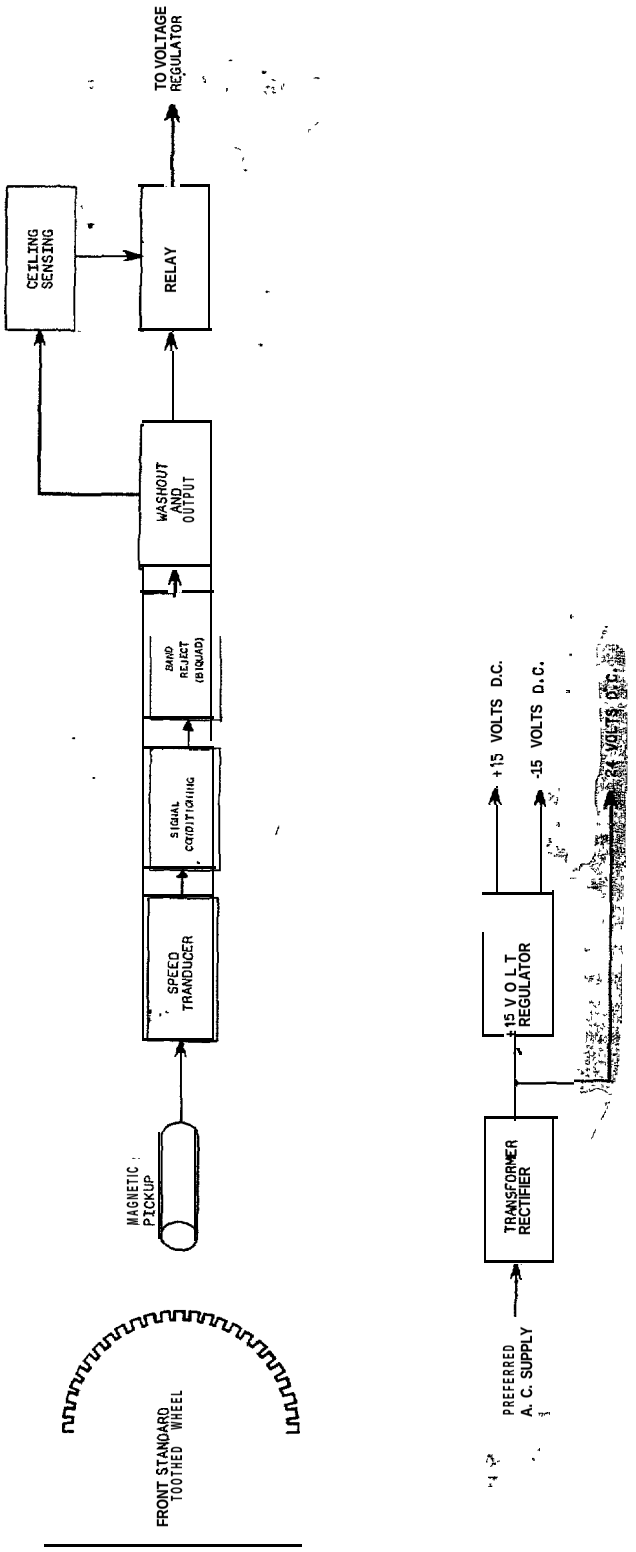


Figure 13, Sheet 1. Power System Stabilizer

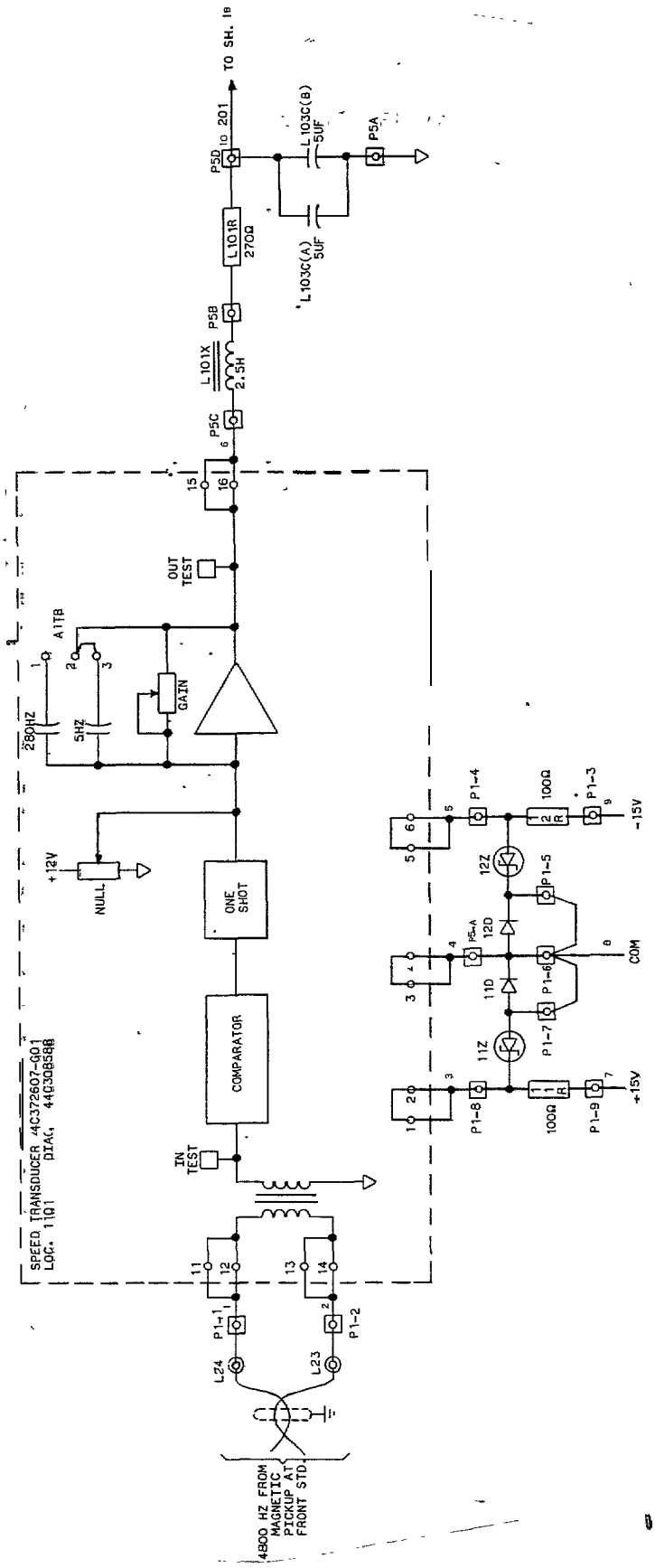


Figure 13, Sheet 1A. Power System Stabilizer

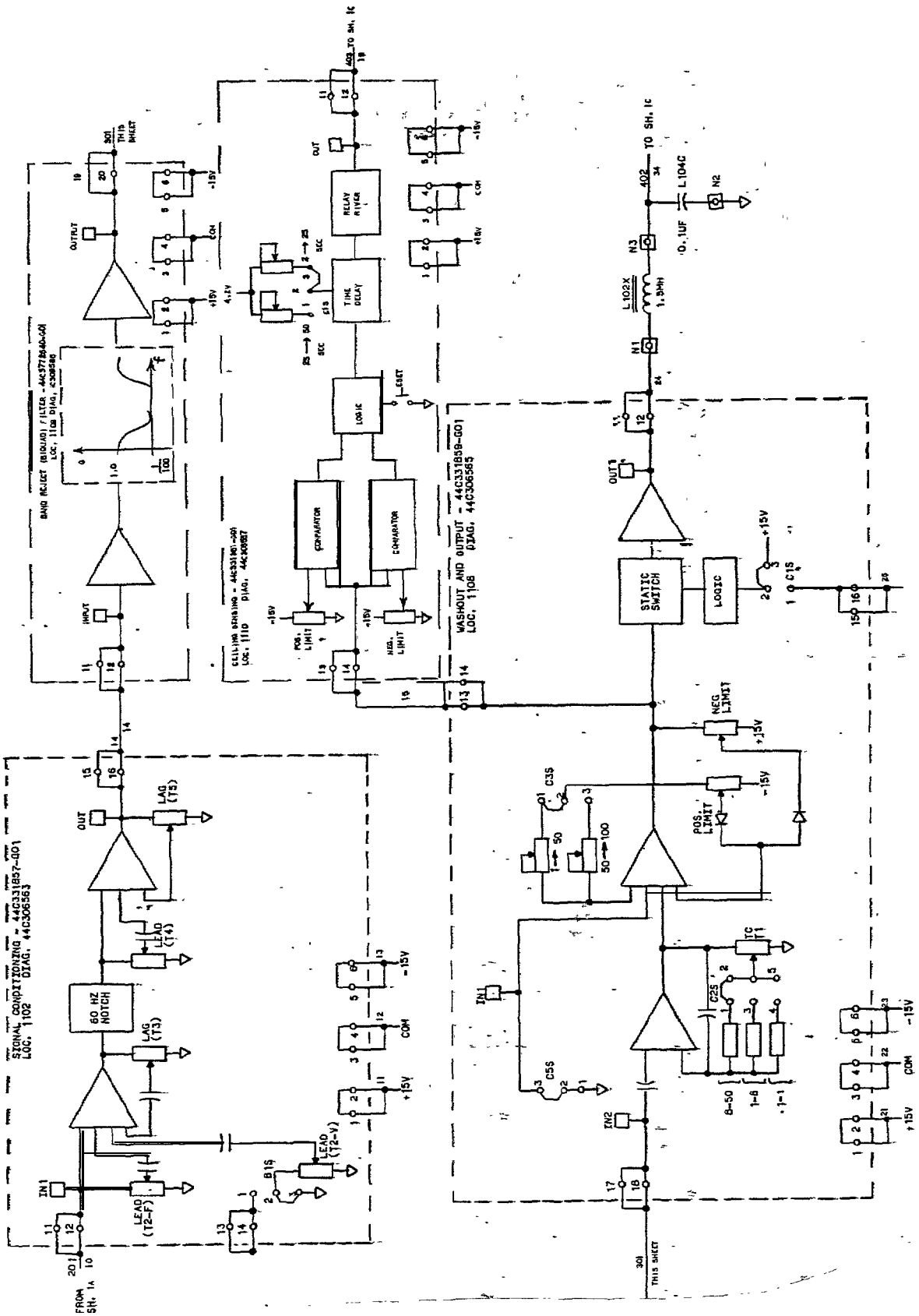


Figure 13, Sheet 1B. Power System Stabilizer

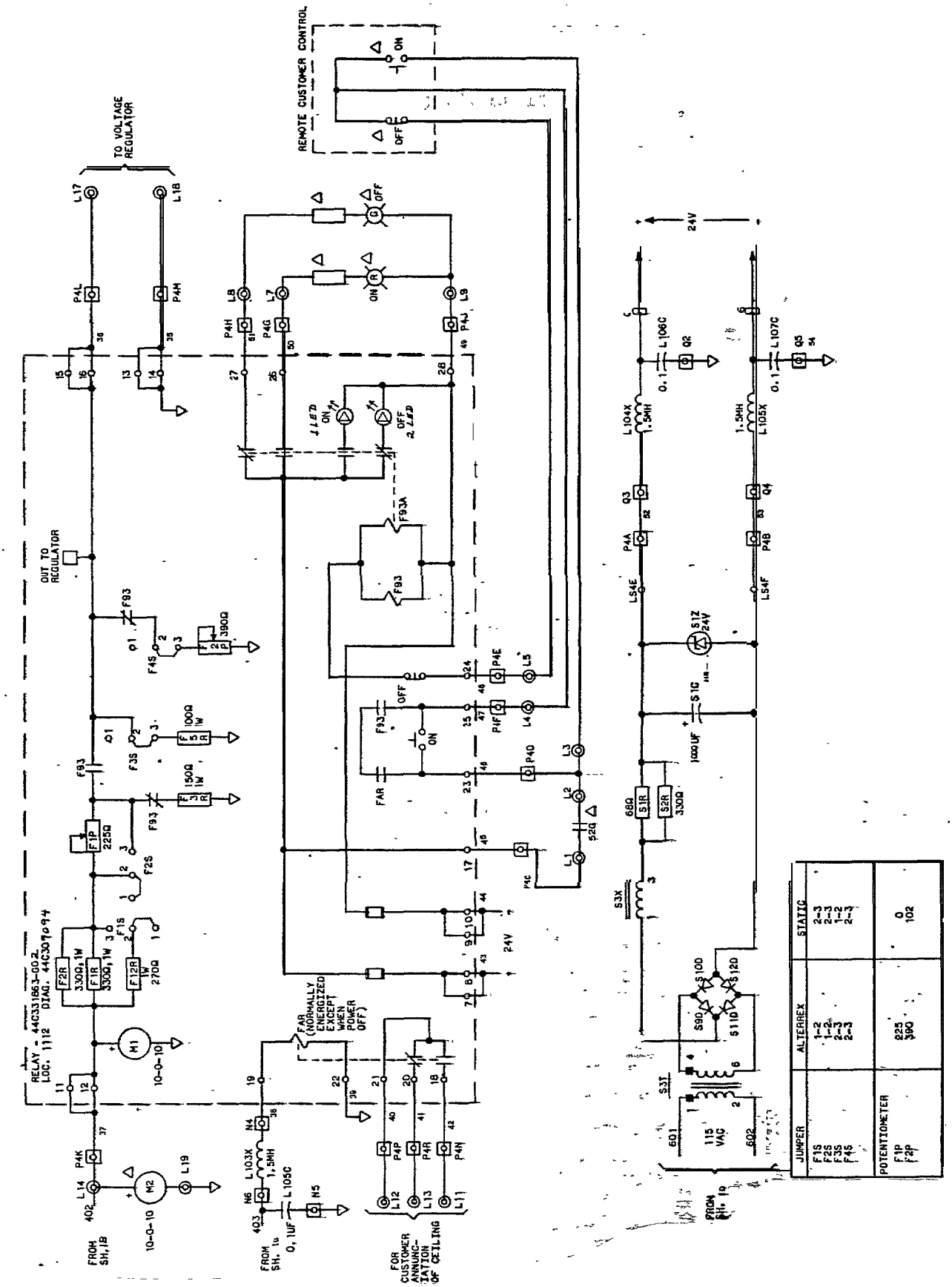


Figure 13, Sheet 1C. Power System Stabilizer

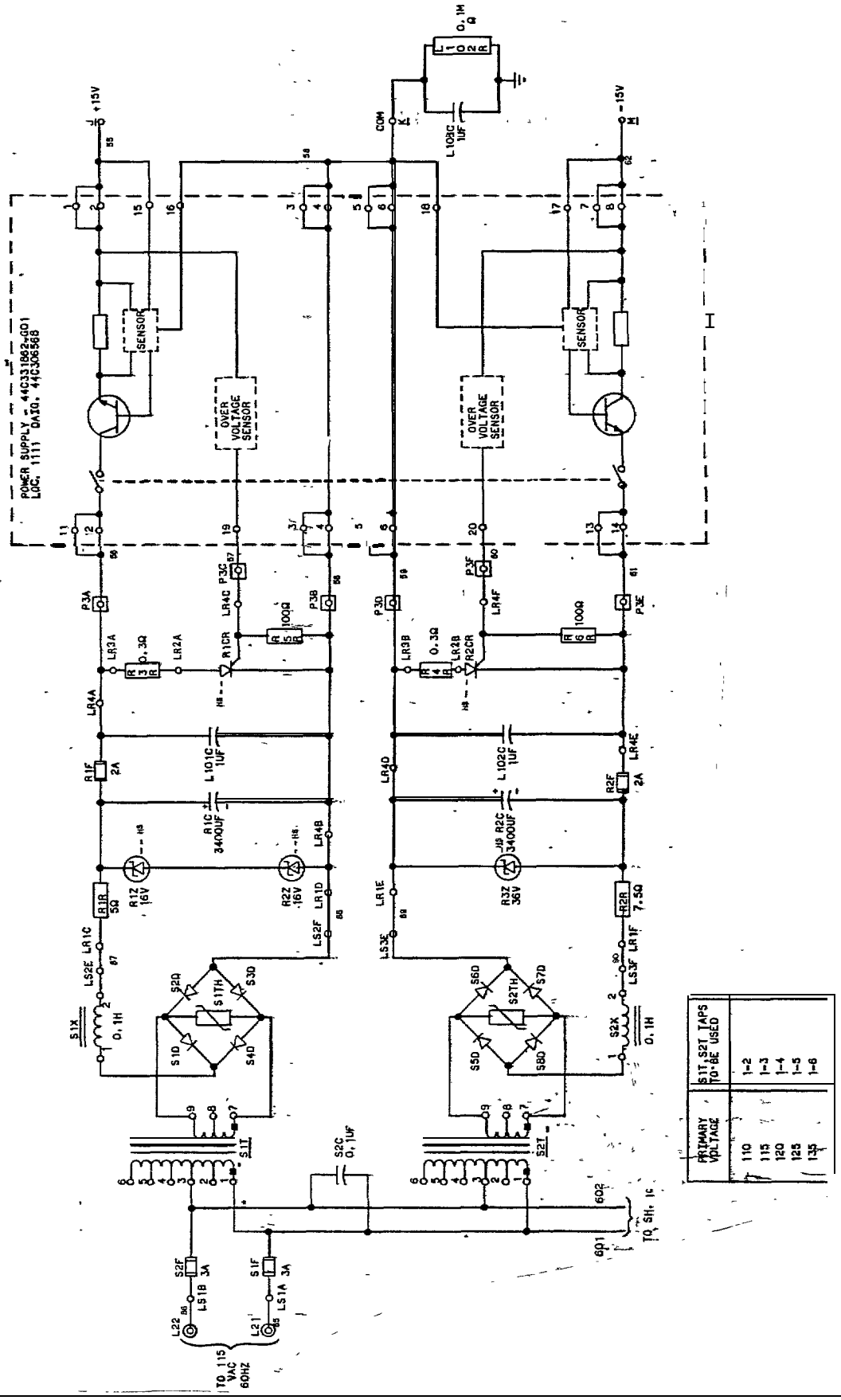
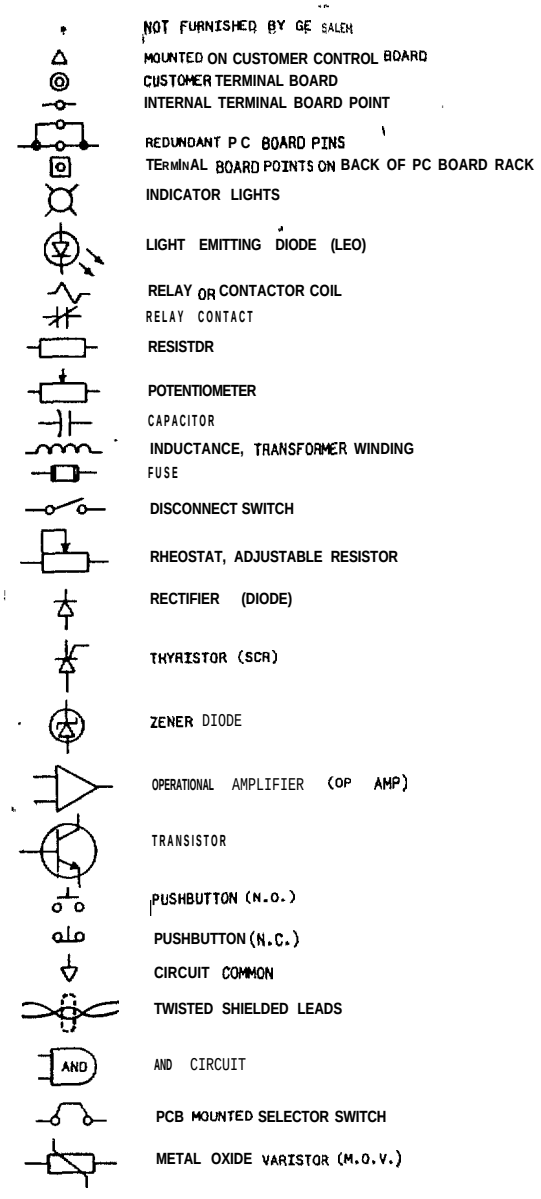


Figure 13, Sheet 1D. Power System Stabilizer



CIRCUIT	SHEET
CEILING SENSING	1B
POWER SUPPLY	
±15 VOLT	1D
24 VOLT	1C
RELAY	1C
SIGNAL CONDITIONING	1B
SPEED TRANSDUCER	1A
WASHOUT AND OUTPUT	1B
BAND REJECT	1B

Figure 13, Sheet 1E. Power System Stabilizer

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